

Roll No.....

SET-II

End Semester Examination, June 2025

Name of the Program: **B.Tech. Biotechnology**

Semester: **VI**

Name of the Course: **Biosensors and Bioelectronics**

Course Code: **TBT604B**

Time: 3 hrs.

Maximum Marks: 100

Note:

(i) Answer **any two questions from each section**

(ii) Each question carries 10 marks

Q1	(20 marks)	
(a)	Define sensors and biosensors. How do they differ? Describe the basic configuration of a biosensor using a block diagram.	CO1, CO2
(b)	Classify biosensors based on the type of bioreceptor used: enzyme, antibody, DNA, and microbial-based biosensors.	
(c)	Explain the working principle of calorimetric biosensors and highlight their key applications. How do these biosensors differ from electrochemical biosensors?	
Q2	(20 marks)	
(a)	Explain the working of nanobiosensors. How do they enhance sensitivity and specificity compared to conventional biosensors?	CO2, CO3
(b)	Discuss in detail the construction and biomedical applications of cochlear implants.	
(c)	Describe the principle and applications of smart bioelectronic devices like Google Smart Lens and their future potential.	
Q3	(20 marks)	
(a)	What are transducers in biosensors? Describe the different types of signal transducers with suitable examples.	CO3, CO4
(b)	Discuss in detail the construction and biomedical applications of cochlear implants.	
(c)	Discuss the key molecular components used in bioelectronics. How do they contribute to the function and performance of bioelectronic devices?	
Q4	(20 marks)	
(a)	Describe the function and application of LFIA-based biosensors in clinical diagnostics. How are antibodies used in these systems?	CO4, CO5
(b)	List and explain the different types of electrodes used in biosensor development. What factors affect electrode performance?	
(c)	Describe the principles and real-world applications of wearable biosensors. How do they facilitate continuous, real-time health monitoring? Also, explain the operational mechanism of the electronic tongue and its role in analytical sensing.	
Q5	(20 marks)	

(a)	Define bioelectronics. Describe the architecture of a bioelectronic system with examples from healthcare devices.	CO5
(b)	How do computing elements like microchips enhance biosensor performance? Give examples from portable diagnostic devices.	
(c)	Discuss the role of Artificial Intelligence (AI) and the Internet of Things (IoT) in the advancement of biosensors. How do these technologies enhance real-time data acquisition, analysis, and decision-making in healthcare monitoring?	